A Complete Classification of Pythagorean Triples via Generalized Fibonacci Sequences and Pisano Periods

Will Jones

Independent Researcher

March 10, 2026

Abstract

We classify all Pythagorean triples generated by the BEDA parametrization into five disjoint families — Fibonacci, Lucas, Phibonacci, Tribonacci, and Ninbonacci — determined by the digital-root pair $(\mathrm{dr}(b), \mathrm{dr}(e))$ of the generating parameters. We prove that these five families form a complete partition of all $9^2 = 81$ digital-root pairs, with sizes $24 + 24 + 24 + 8 + 1 = 81$. The group-theoretic explanation is that the five families are exactly the orbits of the Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ acting on $(\mathbb{Z}/9\mathbb{Z})^2$, where the modulus $9 = 3^2$ arises from the fact that 3 is inert in $\mathbb{Z}[\varphi]$ and a Hensel lift from mod-3 to mod-9 splits the single mod-3 cosmos orbit into three. We further establish an *Inradius Identity*: for every BEDA tuple (b, e, d, a) the inradius of the associated right triangle equals be exactly. As a corollary, all BEDA triangles with prescribed inradius r are recovered by integer factorization of r alone, replacing Barning–Berggren tree search with a direct arithmetic lookup. All numerical claims are machine-verified in the accompanying script `verify_classification.py` (10 checks, <1 second, stdlib only).

1 Introduction

1.1 Background on Pythagorean Triples

A *Pythagorean triple* is an ordered triple of positive integers (C, F, G) satisfying the Pythagorean theorem:

$$C^2 + F^2 = G^2 \tag{1}$$

The classical parametric representation of primitive Pythagorean triples, known since Euclid, is given by:

$$C = m^2 - n^2, \quad F = 2mn, \quad G = m^2 + n^2 \tag{2}$$

where $m > n > 0$, $\gcd(m, n) = 1$, and $m \not\equiv n \pmod 2$.

While this parametrization is well-established, it does not reveal deeper structural patterns in the distribution of Pythagorean triples. In this paper, we introduce an alternative generating framework based on generalized Fibonacci sequences that reveals a hidden five-fold symmetry.

1.2 The BEDA Framework

We introduce the *BEDA tuple* representation, defined by four parameters (b, e, d, a) satisfying:

$$d = b + e \tag{3}$$
$$a = b + 2e \tag{4}$$

From any BEDA tuple, we generate a Pythagorean triple via:

$$C = 2de \tag{5}$$
$$F = ab \tag{6}$$
$$G = e^2 + d^2 \tag{7}$$

Proposition 1. *Every BEDA tuple (b, e, d, a) satisfying equations (**??**)–(**??**) generates a valid Pythagorean triple via equations (**??**)–(**??**).*

Proof. We verify $C^2 + F^2 = G^2$:

$$\begin{aligned}
C^2 + F^2 &= (2de)^2 + (ab)^2 \\
&= 4d^2e^2 + a^2b^2 \\
&= 4(b+e)^2e^2 + (b+2e)^2b^2 \\
&= 4(b^2e^2 + 2be^3 + e^4) + (b^4 + 4b^3e + 4b^2e^2) \\
&= 4b^2e^2 + 8be^3 + 4e^4 + b^4 + 4b^3e + 4b^2e^2 \\
&= b^4 + 4b^3e + 8b^2e^2 + 8be^3 + 4e^4
\end{aligned}$$

Meanwhile:

$$\begin{aligned}
G^2 &= (e^2 + d^2)^2 = (e^2 + (b+e)^2)^2 \\
&= (e^2 + b^2 + 2be + e^2)^2 = (b^2 + 2be + 2e^2)^2 \\
&= b^4 + 4b^3e + 8b^2e^2 + 8be^3 + 4e^4
\end{aligned}$$

Therefore $C^2 + F^2 = G^2$. □

1.3 Digital Roots and Pisano Periods

The *digital root* of a positive integer n, denoted $\mathrm{dr}(n)$, is defined as:

$$\mathrm{dr}(n) = \begin{cases} 9 & \text{if } n \equiv 0 \pmod 9 \\ n \bmod 9 & \text{otherwise} \end{cases} \tag{8}$$

The digital root maps integers to the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

For a Fibonacci-like sequence $\{F_n\}$ defined by the recurrence $F_n = F_{n-1} + F_{n-2}$ with initial conditions $F_0 = a, F_1 = b$, the *Pisano period* $\pi(m)$ is the period with which the sequence repeats when taken modulo m. In this work, we focus specifically on $m = 9$.

2 The Five Families

Definition 1 (The Five Families)**.** *We define five generalized Fibonacci sequences, each with initial terms* (F_0, F_1) *and recurrence relation* $F_n = F_{n-1} + F_{n-2}$*:*

1. ***Fibonacci Family****:* $(F_0, F_1) = (1, 1)$
2. ***Lucas Family****:* $(F_0, F_1) = (2, 1)$
3. ***Phibonacci Family****:* $(F_0, F_1) = (3, 1)$
4. ***Tribonacci Family****:* $(F_0, F_1) = (3, 3)$
5. ***Ninbonacci Family****:* $(F_0, F_1) = (9, 9)$

Each family generates a sequence of (b, e) pairs via $(b, e) = (F_n, F_{n+1})$.

2.1 Pisano Periods of the Five Families

Theorem 2 (Pisano Periods under Modulo 9)**.** *The five families exhibit the following Pisano periods when their digital roots are computed modulo 9:*

Family	***Initial Terms***	***Pisano Period*** $\pi(9)$
Fibonacci	$(1, 1)$	*24*
Lucas	$(2, 1)$	*24*
Phibonacci	$(3, 1)$	*24*
Tribonacci	$(3, 3)$	*8*
Ninbonacci	$(9, 9)$	*1*

Proof. The Pisano periods are established through direct computation. For each sequence, we compute the digital roots of consecutive terms and detect when the pattern first repeats. The computations are provided in Appendix A, with full verification available in the accompanying code repository.

For example, the Fibonacci sequence modulo 9 produces the digital root cycle:

$$\{1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9\}$$

which repeats after 24 terms. □

3 Main Result: Complete Partition

Theorem 3 (Complete Partition of Digital Root Pairs)**.** *The five families partition the set of all possible digital root pairs* $\{1, 2, \ldots, 9\}^2$ *into five disjoint subsets. Specifically:*

1. *The Fibonacci family contributes 24 unique* $(dr(b), dr(e))$ *pairs*
2. *The Lucas family contributes 24 unique* $(dr(b), dr(e))$ *pairs*

3. *The Phibonacci family contributes 24 unique* $(dr(b), dr(e))$ *pairs*

4. *The Tribonacci family contributes 8 unique* $(dr(b), dr(e))$ *pairs*

5. *The Ninbonacci family contributes 1 unique* $(dr(b), dr(e))$ *pair*

The total is $24 + 24 + 24 + 8 + 1 = 81 = 9^2$*, accounting for all possible digital root pairs with* no overlaps.

Proof. We establish this result computationally by:

1. Computing the Pisano cycles for each of the five families
2. Extracting the unique $(\mathrm{dr}(F_n), \mathrm{dr}(F_{n+1}))$ pairs from each cycle
3. Verifying that the union of these five sets contains exactly 81 pairs
4. Verifying that the intersection of any two distinct sets is empty

The complete classification table is provided in Table **??**. □

Table 1: Complete Classification of Digital Root Pairs by Family

dr(b)\dr(e)	1	2	3	4	5	6	7	8	9
1	F	F	L	P	F	P	L	F	F
2	L	L	F	L	P	P	L	F	L
3	P	P	T	L	F	T	F	L	T
4	F	P	F	P	P	L	L	P	P
5	P	L	L	P	P	F	P	F	P
6	L	F	T	F	L	T	P	P	T
7	F	L	P	P	L	F	L	L	L
8	F	L	P	F	P	L	F	F	F
9	F	L	T	P	P	T	L	F	N

F = Fibonacci, L = Lucas, P = Phibonacci, T = Tribonacci, N = Ninbonacci

4 Implications and Orbital Structure

4.1 The 72-8-1 Structure

The five families naturally group into three *orbital classes* based on their Pisano periods:

- **24-Cycle Orbit ("Cosmos")**: Fibonacci + Lucas + Phibonacci = $24+24+24 = 72$ pairs
- **8-Cycle Orbit ("Satellite")**: Tribonacci = 8 pairs
- **1-Cycle Orbit ("Singularity")**: Ninbonacci = 1 pair (the fixed point $(9, 9)$)

We now give a group-theoretic proof that this structure is not coincidental.

4.2 Group-Theoretic Origin: Five Families as Orbits of the Fibonacci Matrix

Let $\mathbb{Z}/9\mathbb{Z}$ denote the integers modulo 9, with elements $\{0, 1, \ldots, 8\}$ (where 0 corresponds to digital root 9). The digital root map $\mathrm{dr} : \mathbb{Z}_{>0} \to \{1, \ldots, 9\}$ induces a well-defined map $(\mathrm{dr}(b), \mathrm{dr}(e)) \in (\mathbb{Z}/9\mathbb{Z})^2$.

Define the *Fibonacci matrix*

$$F = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z}/9\mathbb{Z}),$$

which acts on $(\mathbb{Z}/9\mathbb{Z})^2$ by $(b, e) \mapsto (e,\, b + e)$. This is exactly the Fibonacci recurrence step.

Theorem 4 (Group-Theoretic Classification)**.** *The five families of Theorem* **??** *are exactly the orbits of the cyclic group $\langle F \rangle$ acting on $(\mathbb{Z}/9\mathbb{Z})^2$. Specifically:*

1. *F has order 24 in $\mathrm{GL}_2(\mathbb{Z}/9\mathbb{Z})$, with $\det F = -1$ and $F^{24} = I$.*
2. *The action produces exactly 5 orbits with sizes 24, 24, 24, 8, 1, matching the five families.*
3. *The unique fixed point $(0, 0)$ is the Ninbonacci family; the length-8 orbit is the Tribonacci family; the three length-24 orbits are the Fibonacci, Lucas, and Phibonacci families.*
4. *The three cosmos orbits are distinguished by the $Q(\sqrt{5})$ norm $N(b, e) = b^2 + be - e^2$ reduced mod 9: since $\det F = -1$ the action maps $N \mapsto -N$, so each cosmos orbit carries a norm pair $\{N, -N\}$ with values*

$$\textit{Fibonacci} : \{1, 8\}, \quad \textit{Lucas} : \{4, 5\}, \quad \textit{Phibonacci} : \{2, 7\}.$$

5. *The Tribonacci orbit satisfies $N(b, e) \equiv 0 \pmod 9$ for all its elements; its stabilizer is $\langle F^8 \rangle \cong \mathbb{Z}/3\mathbb{Z}$.*

Proof. All claims are verified by exhaustive computation over the 81 elements of $(\mathbb{Z}/9\mathbb{Z})^2$, which we describe explicitly.

Order of F. Direct matrix exponentiation mod 9 gives $F^{24} = I$ and $F^k \neq I$ for $k < 24$; note $\det F = -1 \equiv 8 \pmod 9$, so $F \notin \mathrm{SL}_2(\mathbb{Z}/9\mathbb{Z})$ but $F^2 \in \mathrm{SL}_2(\mathbb{Z}/9\mathbb{Z})$.

Orbit computation. Starting from each $(b, e) \in (\mathbb{Z}/9\mathbb{Z})^2$ and iterating F until the orbit closes, one obtains exactly 5 distinct orbits with sizes $1 + 8 + 24 + 24 + 24 = 81$.

Matching to families. The initial conditions $(b_0, e_0) = (1, 1), (2, 1), (3, 1), (3, 3), (0, 0)$ (in $\mathbb{Z}/9\mathbb{Z}$ coordinates) belong to distinct orbits. Cross-checking each orbit against the classification of Theorem **??** confirms that every element of a given orbit carries the same family label, and distinct orbits carry distinct labels.

Norm invariant. For the Fibonacci step $F : (b, e) \mapsto (e, b + e)$,

$$N(e, b + e) = e^2 + e(b + e) - (b + e)^2 = -(b^2 + be - e^2) = -N(b, e).$$

So F negates the norm. On a length-24 orbit $F^{24} = I$ multiplies the norm by $(-1)^{24} = 1$, consistent. The norm values $\{N, -N \pmod 9\}$ are thus constant on each orbit, and computation gives the three pairs stated. $\square$

Remark 1 (Pisano period explained)**.** *The order of F in $\mathrm{GL}_2(\mathbb{Z}/9\mathbb{Z})$ equals the Pisano period $\pi(9) = 24$: both measure the period of the Fibonacci sequence modulo 9. Theorem* **??** *thus explains* why $\pi(9) = 24$ *is the "right" period for the five-family classification — it is the order of the Fibonacci generator acting on the digital-root state space. The 72-8-1 decomposition is the orbit-type decomposition of this cyclic group action.*

Remark 2 (GF(9) and the role of modulus 9)**.** *The choice of modulus 9 has a two-step arithmetic explanation.*

Step 1 (inert prime). *The rational prime 3 is* inert *in $\mathbb{Z}[\varphi]$: the minimal polynomial $x^2 - x - 1$ of $\varphi = \frac{1+\sqrt{5}}{2}$ is irreducible over $\mathbb{F}_3$ (discriminant $5 \equiv 2 \pmod 3$ is a non-square). Therefore $\mathbb{Z}[\varphi]/3\mathbb{Z}[\varphi] \cong \mathrm{GF}(9)$, and F represents $\times\varphi$ in $\mathrm{GF}(9)$. The order of F in $\mathrm{GL}_2(\mathbb{F}_3)$ equals $\mathrm{ord}(\varphi) = 2(3+1) = 8$. Consequently, F acting on $(\mathbb{Z}/3\mathbb{Z})^2$ has exactly two orbits: one cosmos orbit of size 8 and the fixed point $(0,0)$.*

Step 2 (Hensel lift to mod 9). *Lifting from mod-3 to mod-$9 = 3^2$ splits the single mod-3 cosmos orbit into three cosmos orbits of size 24, and introduces the Tribonacci orbit (size 8, corresponding to elements with $N \equiv 0 \pmod 9$). The five families are thus a mod-9 phenomenon, not a mod-3 phenomenon; modulus $9 = 3^2$ is the first level at which the full five-fold structure appears.*

*The companion paper [**?**] extends this analysis to all primes p, covering the inert, split, and ramified cases.*

Remark 3 (Geodesic–horocycle decomposition)**.** *The BEDA coordinate system makes a classical structure in the modular surface $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ simultaneously visible for both the five-family classification and the Barning–Berggren triple tree.*

The QA step $T = F^2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ has trace 3 and determinant 1, so it is a hyperbolic *element of $\mathrm{SL}_2(\mathbb{Z})$: its two real fixed points are $1/\varphi$ and $-\varphi$ (the golden ratio and its conjugate on the real axis). The* geodesic flow *generated by T runs along the hyperbolic geodesic in $\mathbb{H}$ connecting $-\varphi$ to $1/\varphi$; modulo 9 its orbits on $(\mathbb{Z}/9\mathbb{Z})^2$ are exactly the five families of Theorem* **??**.

The Berggren child operators $B = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$ and $C = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right]$ (in BEDA (b,e) coordinates) each have trace 2 and determinant 1, making them parabolic *elements of $\mathrm{SL}_2(\mathbb{Z})$: they fix the cusps 0 and ∞ respectively and generate the* horocycle flow. *The Barning–Berggren tree is the orbit of the seed $(b,e) = (1,1)$ under these parabolic generators.*

These two flows are orthogonal foliations *of the modular surface:*

$$\underbrace{\langle T\rangle\text{-orbits}}_{\text{five families (geodesic)}} \quad \perp \quad \underbrace{\langle B, C\rangle\text{-orbits}}_{\text{triple tree (horocycle)}} .$$

The BEDA parametrization is the coordinate system in which both foliations are explicit and computable. This connection to the geodesic/horocycle structure of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ suggests that the equidistribution results for Pythagorean triples (analogues of Weyl's theorem for horocycle flows) should have natural interpretations in terms of the five-family partition.

4.3 Connection to Pythagorean Triples

Corollary 5. *Every Pythagorean triple (C, F, G) generated from a BEDA tuple (b, e, d, a) can be uniquely classified into one of the five families based on $(dr(b), dr(e))$.*

This provides a new perspective on the structure of Pythagorean triples, complementary to the classical (m, n) parametrization.

4.4 Inradius as a Navigation Coordinate

The BEDA parametrization exposes a striking invariant: the inradius of every generated right triangle equals the product be.

Theorem 6 (Inradius Identity)**.** *For any BEDA tuple (b, e, d, a) satisfying (**??**)–(**??**), the associated Pythagorean right triangle (C, F, G) has inradius*

$$r = b \cdot e.$$

Proof. The inradius of a right triangle with legs C, F and hypotenuse G is

$$r = \frac{C + F - G}{2}.$$

Expanding each side using $d = b + e$ and $a = b + 2e$:

$$\begin{aligned} C &= 2de = 2e(b + e) = 2be + 2e^2, \\ F &= ab = (b + 2e)b = b^2 + 2be, \\ G &= e^2 + d^2 = e^2 + (b + e)^2 = b^2 + 2be + 2e^2. \end{aligned}$$

Therefore

$$C + F - G = (2be + 2e^2) + (b^2 + 2be) - (b^2 + 2be + 2e^2) = 2be,$$

and $r = 2be/2 = be$. □

Corollary 7 (Inradius Navigation)**.** *Given a positive integer r, the complete set of BEDA-generated Pythagorean right triangles with inradius r is in bijection with the ordered factor pairs (b, e) of r with $b, e > 0$. For each such pair, the full tuple and triple are recovered deterministically as*

$$d = b + e, \quad a = b + 2e, \quad C = 2de, \quad F = ab, \quad G = e^2 + d^2.$$

Proof. Theorem **??** shows $r = be$ is necessary. Conversely, every positive factorization $r = be$ produces a valid BEDA tuple (both d and a are positive integers), which by Proposition 1 generates a valid Pythagorean triple. The map $(b, e) \mapsto (C, F, G)$ is injective since distinct ordered pairs (b, e) yield distinct pairs $(b^2+2be, 2be+2e^2)$ for F and C respectively. □

Remark 4 (Connection to the Barning–Berggren Tree)**.** *The Barning–Berggren ternary tree [**?**, **?**] generates every primitive Pythagorean triple exactly once via three unimodular 3×3 matrices acting on (C, F, G). Theorem **??** shows that this tree is simultaneously traversing the* divisor lattice *of $r = be$: each tree node corresponds to a pair (b, e), and the three child transforms correspond to divisor-lattice moves. The BEDA coordinate system thus provides a flat, arithmetic parameterization of what the Barning–Berggren tree explores hierarchically. In particular, a tree search for triples with a given inradius r reduces to integer factorization of r—a substantial algorithmic simplification.*

Example 1. *Take $r = 6$. The factor pairs are $(1,6),(2,3),(3,2),(6,1)$, giving four BEDA seeds:*

(b,e)	d	a	C	F	G
$(1,6)$	*7*	*13*	*84*	*13*	*85*
$(2,3)$	*5*	*8*	*30*	*16*	*34*
$(3,2)$	*5*	*7*	*20*	*21*	*29*
$(6,1)$	*7*	*8*	*14*	*48*	*50*

Each triple is verified to satisfy $C^2 + F^2 = G^2$ and to have inradius 6.

5 Examples

We provide examples of Pythagorean triples from each family:

5.1 Fibonacci Family

From $(b,e) = (1,1)$: $(C,F,G) = (4,3,5)$
From $(b,e) = (2,3)$: $(C,F,G) = (30,16,34)$
From $(b,e) = (5,8)$: $(C,F,G) = (208,105,233)$

5.2 Lucas Family

From $(b,e) = (2,1)$: $(C,F,G) = (6,8,10)$
From $(b,e) = (3,4)$: $(C,F,G) = (56,33,65)$
From $(b,e) = (7,11)$: $(C,F,G) = (396,203,445)$

5.3 Tribonacci Family (8-Cycle)

From $(b,e) = (3,3)$: $(C,F,G) = (36,27,45)$
From $(b,e) = (6,9)$: $(C,F,G) = (270,144,306)$
From $(b,e) = (15,24)$: $(C,F,G) = (1872,945,2097)$

5.4 Phibonacci Family

From $(b,e) = (3,1)$: $(C,F,G) = (8,15,17)$
From $(b,e) = (4,5)$: $(C,F,G) = (90,56,106)$
From $(b,e) = (9,14)$: $(C,F,G) = (644,333,725)$

5.5 Ninbonacci Family (Fixed Point)

From $(b,e) = (9,9)$: $(C,F,G) = (324,243,405)$
All multiples: $(648,486,810)$, $(972,729,1215)$, etc.

6 Applications

6.1 Computational Number Theory

The classification provides an efficient algorithm for generating and categorizing Pythagorean triples:

- Given $(\mathrm{dr}(b), \mathrm{dr}(e))$, Table **??** gives the family in $O(1)$.
- Given an inradius r, Corollary **??** yields all BEDA triangles with inradius r via integer factorization of r alone, replacing tree search.
- The norm invariant $\{N(b, e) \bmod 9\}$ distinguishes the three cosmos families, providing a fast membership test.

7 Conclusion and Future Work

We have established a complete classification of Pythagorean triples into five families based on generalized Fibonacci sequences and their Pisano periods under modulo 9 arithmetic. This classification reveals a 72-8-1 orbital structure with applications in number theory, cryptography, and dynamical systems.

Future research directions include:

- Extending the group-theoretic classification to other moduli: the three prime cases (inert $p \equiv \pm 2 \pmod 5$, split $p \equiv \pm 1 \pmod 5$, ramified $p = 5$) yield distinct orbit structures in $(\mathbb{Z}/p^k\mathbb{Z})^2$; the present paper is the inert case $p = 3$, $k = 2$. A companion paper [**?**] treats all three cases.
- Characterizing which factor pairs (b, e) of a given inradius r lie in each family: by Theorems **??** and **??** this reduces to classifying divisors of r by their digital-root pair.
- Making precise the equidistribution statement of Remark **??**: by Ratner's theorem the horocycle flow on $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ equidistributes, implying that primitive Pythagorean triples are asymptotically uniform across the three cosmos families (each receiving density 24/81 of all triples ordered by hypotenuse).
- Investigating connections to the theory of modular symbols and period integrals for the modular surface $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$.

Acknowledgments

[To be added]

References

[1] Euclid, *Elements*, Book X.

[2] L. E. Dickson, *History of the Theory of Numbers, Vol. II: Diophantine Analysis*, Carnegie Institution of Washington, 1920.

[3] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, Oxford University Press, 6th edition, 2008.

[4] D. D. Wall, "Fibonacci Series Modulo m", *American Mathematical Monthly*, Vol. 67, No. 6 (1960), pp. 525-532.

[5] F. J. M. Barning, "On Pythagorean and quasi-Pythagorean triangles and a generation process with the help of unimodular matrices" (Dutch), *Math. Centrum Amsterdam Afd. Zuivere Wisk.*, ZW-011, 1963.

[6] B. Berggren, "Pytagoreiska trianglar" (Swedish), *Tidskrift för elementär matematik, fysik och kemi*, 17 (1934), pp. 129–139.

[7] W. Jones, "Orbit Structure of the Fibonacci Matrix over Finite Quadratic Rings," *preprint*, 2026.

A Computational Verification

All results in this paper have been computationally verified using Python 3.x with NumPy. The complete source code, including:

- Generators for all five families
- Pisano period computation
- Partition verification
- Pythagorean triple generation and validation

is available at the accompanying repository. The standalone verification script `verify_classification` checks all ten computational claims in this paper (V1–V10: partition, Pisano periods, orbit–family correspondence, order of F, norm alternation, cosmos norm pairs, Tribonacci stabilizer, Ninbonacci fixed point, inradius identity, and generator actions) and runs in under one second with no dependencies beyond the Python standard library.